

Creation Date 22-Dec-2009

Revision Date 08-Feb-2024

Revision Number 3

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                           |
|---------------------------|---------------------------|
| Product Description:      | <b>Arsenic(III) oxide</b> |
| Cat No. :                 | <b>11173</b>              |
| Synonyms                  | Arsenic trioxide          |
| Index No                  | 033-003-00-0              |
| CAS No                    | 1327-53-3                 |
| EC No                     | 215-481-4                 |
| Molecular Formula         | O3 As2                    |
| REACH registration number | -                         |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recommended Use      | Laboratory chemicals. REACH (1907/2006) - Annex XIV. Article 56 (3) REACH. Scientific research and development. The substance is used under strictly controlled conditions. |
| Uses advised against | All other uses                                                                                                                                                              |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

ALFAA11173

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Based on available data, the classification criteria are not met

### Health hazards

Acute oral toxicity  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Carcinogenicity

Category 2 (H300)  
Category 1 (H314) B  
Category 1 (H318)  
Category 1A (H350)

### Environmental hazards

Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1 (H400)  
Category 1 (H410)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### **Hazard Statements**

H300 - Fatal if swallowed  
H314 - Causes severe skin burns and eye damage  
H350 - May cause cancer  
H410 - Very toxic to aquatic life with long lasting effects

### **Precautionary Statements**

P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor/physician

### **Additional EU labelling**

Restricted to professional users

## 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

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## 3.1. Substances

| Component                                       | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                         |
|-------------------------------------------------|-----------|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | 1327-53-3 | EEC No. 215-481-4 | <=100    | Acute Tox. 2 (H300)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Carc. 1A (H350)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) |

| Component                                       | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-------------------------------------------------|---------------------------------------|----------|-----------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | -                                     | 1        | -               |

### Note

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

|                           |   |
|---------------------------|---|
| REACH registration number | - |
|---------------------------|---|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| Eye Contact                        | In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                              |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| Ingestion                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Notes to Physician | Treat symptomatically. |
|--------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

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## 5.1. Extinguishing media

### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

## 5.2. Special hazards arising from the substance or mixture

The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

### **Hazardous Combustion Products**

arsenic oxides.

## 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid dust formation.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe (dust, vapor, mist, gas). Avoid dust formation.

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1B

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Switzerland - Storage of hazardous substances

Storage class - SC 6.1

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020.

**CH** - The Government of Switzerland

has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component                                       | European Union | The United Kingdom                                                                           | France                                       | Belgium | Spain                                          |
|-------------------------------------------------|----------------|----------------------------------------------------------------------------------------------|----------------------------------------------|---------|------------------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) |                | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Carc. except Arsine | TWA / VME: 0.2 mg/m <sup>3</sup> (8 heures). |         | TWA / VLA-ED: 0.01 mg/m <sup>3</sup> (8 horas) |

| Component                                       | Italy | Germany | Portugal                            | The Netherlands                      | Finland                                |
|-------------------------------------------------|-------|---------|-------------------------------------|--------------------------------------|----------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) |       | Haut    | TWA: 0.01 mg/m <sup>3</sup> 8 horas | TWA: 0.0028 mg/m <sup>3</sup> 8 uren | TWA: 0.01 mg/m <sup>3</sup> 8 tunteina |

| Component                                       | Austria                        | Denmark | Switzerland                                        | Poland | Norway                                      |
|-------------------------------------------------|--------------------------------|---------|----------------------------------------------------|--------|---------------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | TRK-TMW: 0.1 mg/m <sup>3</sup> |         | Haut/Peau<br>TWA: 0.01 mg/m <sup>3</sup> 8 Stunden |        | TWA: 0.005 mg/m <sup>3</sup> 8 timer<br>Hud |

| Component                                       | Bulgaria | Croatia                                     | Ireland | Cyprus | Czech Republic |
|-------------------------------------------------|----------|---------------------------------------------|---------|--------|----------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) |          | TWA-GVI: 0.1 mg/m <sup>3</sup> 8 satima. As |         |        |                |

| Component                                       | Russia | Slovak Republic                                                                  | Slovenia                                                                                                          | Sweden | Turkey |
|-------------------------------------------------|--------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) |        | TWA: 0.1 mg/m <sup>3</sup> 8 hodinách<br>STEL: 0.5 mg/m <sup>3</sup> 15 minútach | TWA: 0.1 mg/m <sup>3</sup> 8 urah inhalable fraction<br>STEL: 0.4 mg/m <sup>3</sup> 15 minutah inhalable fraction |        |        |

#### Biological limit values

List source(s):

| Component                                       | European Union | United Kingdom | France                                                                       | Spain | Germany |
|-------------------------------------------------|----------------|----------------|------------------------------------------------------------------------------|-------|---------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) |                |                | Metabolites of inorganic Arsenic: 0.05 mg/g creatinine urine end of workweek |       |         |

#### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

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See table for values

| Component                                                             | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-----------------------------------------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> )<br>1327-53-3 ( ≤100 ) |                              |                                 |                                | DNEL = 112µg/kg bw/day            |

| Component                                                             | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-----------------------------------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> )<br>1327-53-3 ( ≤100 ) |                                  |                                     |                                    | DNEL = 5µg/m <sup>3</sup>             |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                                             | Fresh water     | Fresh water sediment          | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)      |
|-----------------------------------------------------------------------|-----------------|-------------------------------|--------------------|------------------------------------|-------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> )<br>1327-53-3 ( ≤100 ) | PNEC = 17.1µg/L | PNEC = 171.1mg/kg sediment dw | PNEC = 1.2µg/L     | PNEC = 80.3µg/L                    | PNEC = 0.7mg/kg soil dw |

| Component                                                             | Marine water   | Marine water sediment      | Marine water Intermittent | Food chain            | Air |
|-----------------------------------------------------------------------|----------------|----------------------------|---------------------------|-----------------------|-----|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> )<br>1327-53-3 ( ≤100 ) | PNEC = 1.2µg/L | PNEC = 12mg/kg sediment dw |                           | PNEC = 1.31mg/kg food |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure

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limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Solid                    |                                   |
| Appearance                              | Beige                    |                                   |
| Odor                                    | Odorless                 |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | 312.3 °C / 594.1 °F      |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flammability (liquid)                   | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | Not applicable           |                                   |
| Viscosity                               | Not applicable           | Solid                             |
| Water Solubility                        | 37 g/L (20°C)            |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Arsenic oxide (As2O3)                   | 18.1                     |                                   |
| Vapor Pressure                          | 66 mmHg @ 312 °C         |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Particle characteristics                | No data available        |                                   |

### 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | O3 As2                 |
| Molecular Weight  | 197.84                 |
| Evaporation Rate  | Not applicable - Solid |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Avoid dust formation.

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## 10.5. Incompatible materials

Strong oxidizing agents.

## 10.6. Hazardous decomposition products

arsenic oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Category 2

Dermal

No data available

Inhalation

No data available

| Component                                       | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|-------------------------------------------------|-------------------------|-------------|-----------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | LD50 = 20 mg/kg ( Rat ) | -           | -               |

##### (b) skin corrosion/irritation;

Category 1 B

##### (c) serious eye damage/irritation;

Category 1

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

##### (e) germ cell mutagenicity;

No data available

##### (f) carcinogenicity;

Category 1A

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component                                       | EU           | UK | Germany | IARC    |
|-------------------------------------------------|--------------|----|---------|---------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | Carc Cat. 1A |    | Cat. 1  | Group 1 |

##### (g) reproductive toxicity;

No data available

##### (h) STOT-single exposure;

No data available

##### (i) STOT-repeated exposure;

No data available

Target Organs

None known.

##### (j) aspiration hazard;

Not applicable  
Solid

#### Symptoms / effects, both acute and delayed

Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated.

### 11.2. Information on other hazards



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**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component                                       | Freshwater Fish                                                                                                                                                                  | Water Flea                                                             | Freshwater Algae |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | LC50: = 135 mg/L, 96h<br>(Pimephales promelas)<br>LC50: > 1000 mg/L, 96h static<br>(Oncorhynchus mykiss)<br>LC50: 18.8 - 21.4 mg/L, 96h<br>flow-through (Oncorhynchus<br>mykiss) | EC50 = 0.038 mg/L 24h<br>EC50 = 0.96 mg/L 96h<br>EC50 = 0.038 mg/L 24h |                  |

| Component                                       | Microtox                                                                                                    | M-Factor |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | EC50 = 31.43 mg/L 60 min<br>EC50 = 33.39 mg/L 30 min<br>EC50 = 43.56 mg/L 15 min<br>EC50 = 73.73 mg/L 5 min | 1        |

### 12.2. Persistence and degradability

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary

#### Persistence

#### Degradability

#### Degradation in sewage treatment plant

May persist.

Not relevant for inorganic substances.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

Product has a high potential to bioconcentrate

| Component                                       | log Pow | Bioconcentration factor (BCF) |
|-------------------------------------------------|---------|-------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | 18.1    | 80 - 236 dimensionless        |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles

### 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant

This product does not contain any known or suspected substance

#### Ozone Depletion Potential

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

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|                                            |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                        |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                  |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.                                              |
| <b>Switzerland - Waste Ordinance</b>       | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                         |                  |
|-----------------------------------------|------------------|
| <b>14.1. UN number</b>                  | UN1561           |
| <b>14.2. UN proper shipping name</b>    | ARSENIC TRIOXIDE |
| <b>14.3. Transport hazard class(es)</b> | 6.1              |
| <b>14.4. Packing group</b>              | II               |

### ADR

|                                         |                  |
|-----------------------------------------|------------------|
| <b>14.1. UN number</b>                  | UN1561           |
| <b>14.2. UN proper shipping name</b>    | ARSENIC TRIOXIDE |
| <b>14.3. Transport hazard class(es)</b> | 6.1              |
| <b>14.4. Packing group</b>              | II               |

### IATA

|                                         |                  |
|-----------------------------------------|------------------|
| <b>14.1. UN number</b>                  | UN1561           |
| <b>14.2. UN proper shipping name</b>    | ARSENIC TRIOXIDE |
| <b>14.3. Transport hazard class(es)</b> | 6.1              |
| <b>14.4. Packing group</b>              | II               |

|                                                                      |                                                                                                          |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>14.5. Environmental hazards</b>                                   | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
| <b>14.6. Special precautions for user</b>                            | No special precautions required.                                                                         |
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods                                                                           |

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No | EINECS | ELINCS | NLP | IECSC | TCSI | KECL | ENCS | ISHL |
|-----------|--------|--------|--------|-----|-------|------|------|------|------|
|-----------|--------|--------|--------|-----|-------|------|------|------|------|

# SAFETY DATA SHEET

Arsenic(III) oxide

Revision Date 08-Feb-2024

|                       |           |           |                                               |   |     |      |          |       |       |
|-----------------------|-----------|-----------|-----------------------------------------------|---|-----|------|----------|-------|-------|
| Arsenic oxide (As2O3) | 1327-53-3 | 215-481-4 | -                                             | - | X   | X    | KE-09858 | X     | X     |
|                       |           |           |                                               |   |     |      |          |       |       |
| Component             | CAS No    | TSCA      | TSCA Inventory notification - Active-Inactive |   | DSL | NDSL | AICS     | NZIoC | PICCS |
| Arsenic oxide (As2O3) | 1327-53-3 | X         | ACTIVE                                        |   | X   | -    | X        | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component                                       | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization                                                          | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                                                                     | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | 1327-53-3 | Carcinogenic Category 1A, Article 57<br>Application date: November 21, 2013<br>Sunset date: May 21, 2015<br>Exemption - None | Use restricted. See item 72.<br>(see link for restriction details)<br>Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) Use restricted. See item 19.<br>(see link for restriction details) | SVHC Candidate list - 215-481-4 - Carcinogenic, Article 57a                                           |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

### REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Seveso III Directive (2012/18/EC)

| Component                                       | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | 1327-53-3 | Not applicable                                                                            | 0.1 tonne                                                                                |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

| Component                                                             | ANNEX I - PART 1<br>List of chemicals subject to export notification procedure (referred to in Article 8) | ANNEX I - PART 2<br>List of chemicals qualifying for PIC notification (referred to in Article 11) | ANNEX I - PART 3<br>List of chemicals subject to the PIC procedure (referred to in Articles 13 and 14) |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> )<br>1327-53-3 ( ≤100 ) | p(2) — other pesticide including biocides<br>sr — severe restriction                                      | -                                                                                                 | -                                                                                                      |

<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32012R0649&qid=1604065742303>.

### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at

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work .

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component                                       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-------------------------------------------------|---------------------------------------|-------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | WGK3                                  |                         |

| Component                                       | France - INRS (Tables of occupational diseases)               |
|-------------------------------------------------|---------------------------------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> ) | Tableaux des maladies professionnelles (TMP) - RG 20,RG 20bis |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                                                             | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Arsenic oxide (As <sub>2</sub> O <sub>3</sub> )<br>1327-53-3 ( ≤100 ) | Prohibited and Restricted Substances                                                                           |                                                                                 | Annex I - pesticide                                                                         |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H300 - Fatal if swallowed

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

H350 - May cause cancer

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

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OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical incident response training.

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

First aid for chemical exposure, including the use of eye wash and safety showers.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

## Prepared By

Health, Safety and Environmental Department

## Creation Date

22-Dec-2009

## Revision Date

08-Feb-2024

## Revision Summary

New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**